

Safety Belt System



WARNING: All seatbelt components must be inspected and corrected as part of any collision repair. Inspect all seatbelt components as prescribed by Seatbelt Procedure After a Collision found in Section 501-20A General Procedures of this manual. Failure to follow this instruction may result in incorrect operation of the seatbelt system and increase the risk of serious personal injury or death in a crash.

When installing new active occupant restraint components, use only replacement parts specified in the Ford Catalog Advantage™ or equivalent.

The active restraint system consists of the following serviceable items:

- Front safety belt buckles, attached to the inboard side of the driver and passenger seat tracks
- Front center safety belt retractor attached to the RH side of the front center seat back frame
- Front center safety belt buckle attached to the LH side of the front center seat base frame
- Detachable safety belt anchor attached to the RH side front center seat base frame
- Front safety belt retractors and pretensioners, located behind the B-pillar trim panels (Crew Cab and regular cab) or in the rear door (SuperCab)
- Belt Tension Sensor (BTS) , located at the safety belt anchor point of the front passenger outboard safety belt
- Rear outboard safety belt retractors, located behind the B-pillar trim panels (SuperCab) or C-pillar trim panels (Crew Cab)
- Rear center safety belt retractor attached to the back panel of the cab
- RH rear safety belt buckles, attached to the floor pan (SuperCab)
- RH rear safety belt buckles, seat mounted and attached to the seat risers (Crew Cab)
- LH rear safety belt buckles attached to the floor pan (SuperCab and Crew Cab)
- Rear child safety seat tether anchors, at all seating positions, attached to the back panel of the cab
- Lower Anchors and Tethers for CHildren (LATCH) attached to the rear seat risers (Crew Cab and SuperCab only)

Safety Belt Retractors

The safety belt retractors include the following features:

- Dual locking mode
 - Emergency Locking Retractor (ELR)
 - Automatic Locking Retractor (ALR) (all except driver seat, front center seat, and rear center seat)
- Safety belt retractors and pretensioners (front outboard only)
- BTS (front outboard passenger only)
- Cinch tongue (front center and rear center seats only)

Automatic Locking Retractor (ALR)

The ALR mode is used when locking a child seat in an outboard seating position. The ALR mode is automatically activated when the safety belt webbing is fully extracted from the retractor and then allowed to retract. As the safety belt webbing is retracted back onto the spool, an audible clicking sound is made indicating the safety belt retractor is in ALR mode and the safety belt webbing will not pull back out of the safety belt retractor. To disengage the ALR mode, allow the safety belt webbing to fully retract back onto the spool. The ALR mode is disengaged when the webbing is free to extract and retract back into the retractor. For testing, refer to the appropriate Functional Test procedure in Safety Belt System in this section.

Cinch Tongue

Front center and rear center safety belt retractors are equipped with a locking cinch tongue. The locking cinch tongue slides up and down the safety belt webbing when it is in the stowed position or while putting the safety belt on. When the locking

cinch tongue is latched into the buckle, the cinch tongue allows the lap portion of the safety belt to become shorter, but locks the webbing in place to restrict it from becoming longer. Before latching a cinch tongue into the buckle, the lap portion of the safety belt may have to be lengthened. To lengthen the lap belt, pull some webbing out of the safety belt retractor. While holding the webbing below the tongue, grasp the tip (metal portion) of the tongue so that it is parallel to the webbing and slide the tongue upward. Provide enough lap belt length so the tongue can reach the buckle. To fasten a cinch tongue, pull the safety belt from the retractor so the shoulder belt portion of the safety belt crosses the occupant's shoulder and chest. Be sure the belt is not twisted. If the belt is twisted, remove the twist. For instructions on how to remove a twist, refer to Safety Belt Tongue Rotated on Belt in this section. Insert the locking cinch tongue into the correct buckle for the seating position until a snap is heard and latching is felt. Make sure the tongue is securely fastened to the buckle by pulling on the tongue. Adjust the lap belt portion of the safety belt by pulling up on the shoulder belt until the lap belt fits snugly and as low as possible around the occupant's hips.

Emergency Locking Retractor (ELR)

The **ELR** is a vehicle-sensitive feature designed to activate and lock the safety belt webbing during hard braking, hard cornering or in an impact of approximately 24 km/h (15 mph). The **ELR** feature helps to reduce the forward movement of the driver and passengers. For testing, refer to the appropriate Functional Test procedure in Safety Belt System in this section.

Safety Belt Retractor Pretensioners

The pretensioner is a pyrotechnic device that deploys when activated by the Restraints Control Module (RCM) to remove excess safety belt webbing from the shoulder and lap safety belt in the event of an impact. If a deployment of the safety belt retractor pretensioners occurs, a new assembly must be installed. Refer to Safety Belt Retractor and Pretensioner — Regular Cab and Crew Cab or Safety Belt Retractor and Pretensioner — SuperCab in this section.

For safety belt retractor pretensioner diagnostic and disposal information, refer to Section 501-20B.

Belt Tension Sensor (BTS)

The **BTS** is located at the RH front safety belt retractor pretensioner anchor point and operates as part of the Occupant Classification System (OCS) system. To diagnose the **BTS**, refer to Section 501-20B.

Safety Belt Buckles

Front Safety Belt Buckles

The safety belt buckles are attached to the inboard side of the driver and passenger seat tracks and, if equipped, to the LH side of the center seat base frame.

After any crash, safety belt buckles must be inspected to verify correct function.

Rear Safety Belt Buckles

On the SuperCab, the RH rear safety belt buckles are attached to the floor and the LH rear safety belt buckle is part of the rear center safety belt retractor and attaches to the floor. On the Crew Cab, the RH rear safety belt buckles are attached to the seat risers and the LH rear safety belt buckle is part of the rear center safety belt retractor and attaches to the floor.

After any crash, safety belt system components (including retractors and buckles) must be inspected to verify correct function.

Child Safety Seat Restraints

The child safety seat restraints include the following items:

- Serviceable rear child safety seat tether anchors, at all seating positions, attached to the back panel of the cab.
- Non-serviceable Lower Anchors and Tethers for CHildren attached to the rear seat risers on the SuperCab and Crew Cab only. Regular cab vehicles are not equipped with Lower Anchors and Tethers for CHildren.

If a child safety seat was in use during a collision, inspect the child safety seat component mounting areas and repair any damage and restore the vehicle to the original production configuration.